

# Mega XL — custom upgrades & build notes

*Upgrades to a Kickstarter v1.0 Mega XL CNC router.*

This documents the changes made to a first-generation (Kickstarter v1.0) Mega XL: a taller, stiffer Z, closed-loop motors, all-proximity homing, and a complete electronics rebuild around a grblHAL controller in a custom enclosure with a proper 48 V power system, air assist, and probing.

## At a glance

| Subsystem             | Stock (v1.0)                 | Upgraded                                                            |
|-----------------------|------------------------------|---------------------------------------------------------------------|
| Vertical clearance    | stock                        | +3" via Mark Smith riser kit                                        |
| Z assembly            | stock Z                      | Mark Smith Z gantry (only V-wheels + X motor mount reused)          |
| Spindle               | DeWalt DWP611 (1¼ HP router) | 2.2 kW (3 HP) Ø80 mm water-cooled, ER20, 24k RPM, VFD-driven        |
| Stepper motors        | generic open-loop steppers   | all-in-one closed-loop steppers (driver integrated in motor)        |
| Limit / homing        | mechanical limit switches    | proximity sensors on every axis                                     |
| Probing               | —                            | touch plate + 3D probe in parallel; grounded spindle (no tool clip) |
| Controller            | 8-bit Arduino board          | grblHAL Teensy 4.x (USB + Ethernet)                                 |
| Electronics enclosure | cramped stock steel box      | custom 12×16×6" aluminum / plexiglass                               |
| Power                 | stock                        | 48 V Meanwell system, E-stop, buck-derived 12 V / 5 V rails         |
| Air assist            | none                         | Mist-relay 12 V solenoid → 30 PSI nozzle at the cut                 |
| Drag chains           | stock 1.5                    | Y axis upsized to 2" to fit the full cable bundle                   |
| Dust collection       | —                            | 4" swivel + 5' flex to a spindle dust shoe, on an overhead reel     |

## Frame & mechanical

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- **Mark Smith riser kit** — adds **3" of additional vertical (Z) clearance**, allowing taller stock and longer tooling.
- **Mark Smith Z gantry** — a **complete replacement of the Z assembly**. The only stock parts carried over are the **four V-wheels** and the **X stepper motor mount**; everything else is new.

## Spindle & cooling

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The stock **DeWalt DWP611** (1¼ HP trim router) was replaced with a **2.2 kW (3 HP) water-cooled spindle** kit:

- **Spindle** — Ø80 mm body, 220 V, water-cooled, **ER20 collet**, up to **24,000 RPM at 400 Hz**.
- **VFD** — 2.2 kW (3 HP) inverter, fed from the **single-phase 220 V** circuit through the E-stop; it generates the **3-phase output to the spindle** and is commanded by the breakout board's **0–10 V PWM** speed signal (see Power distribution).
- **Mount** — 80 mm spindle clamp on the new Z gantry.
- **Braking resistor** — a 150 W resistor on the VFD dissipates regen energy for a fast stop (spindle down in **under 6 s**).

**Coolant loop.** A pump **submerged in a 1-gallon reservoir** circulates coolant through the spindle via **two 8 mm tubes** (supply and return): *reservoir* → *pump* → *spindle* → *radiator* → *reservoir*. The radiator is cooled by the 12 V fan (see the power tree). The pump is **interlocked to the spindle** — a **VFD digital output** drives the **NO contact of a 110 V relay** the pump plugs into — so it runs whenever the spindle does and the spindle can't be run dry. (Separate from the breakout board's Mist relay, which handles air assist.) The coolant tubes, the air-assist line, and dust extraction all reach the spindle through the overhead umbilical (below).

## Motors

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The generic steppers that shipped with the Mega XL were replaced with **all-in-one closed-loop stepper motors** — each motor has its stepper driver built into the motor body. This removes the separate driver boards, adds closed-loop position feedback (lost-step detection / alarm), and simplifies wiring to a single cable per motor.

- **Individual enable** per motor — each stepper's **EN** line is driven separately.
- The motors' **AL (alarm) outputs are wire-OR'd together** and fed to the controller's **ST0 input**, so a stall / lost-step fault on any motor signals the controller.

All four axes are closed-loop. The machine is a **dual-Y gantry** — the two Y gantries are driven by ganged motors (**Y + A**), one per side.

| Axis  | Holding torque | Drives                                                |
|-------|----------------|-------------------------------------------------------|
| Y + A | 3.2 N·m each   | the two Y gantries (ganged)                           |
| X     | 2.0 N·m        | spindle carriage along the cross-beam (rack & pinion) |
| Z     | 1.2 N·m        | Z                                                     |

The breakout board also provides a **5th axis (B)**. No rotary is fitted yet, but its **step/dir/enable is wired through to the extension panel** so one can be added later by plugging it in — see [Back-of-enclosure harness & connectors](#).

All motors have an **8 mm output shaft with a flat**, with pinions/couplings retained by a grub screw on the flat. *Build note:* the stock Mega XL pinions were 6 mm-bore and were **reamed to 8 mm** at a machine shop — this thinned the set-screw boss and reduced grub-screw thread engagement. See the [rack-and-pinion maintenance](#) appendix.

## Homing & limit sensing

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The Mark Smith Z gantry uses a **proximity sensor** as its limit switch. To keep one sensing scheme across the machine, **all remaining stock mechanical limit switches were replaced with the same style of proximity sensor**, so every axis homes on identical, non-contact hardware.

## Probing

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- The **touch-plate input is wired in parallel with the primary probe input**, alongside the **3D mechanical probe** — either device can trigger the same probe input.
- The **spindle mount is grounded**, so the tool itself is at ground through the spindle. That completes the touch-plate circuit on contact — **no alligator clip on the tool** is required.

## Controller & electronics enclosure

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- Discarded the **cramped stock steel electronics enclosure** and the **stock 8-bit Arduino controller board**.
- Control is now a **grblHAL Teensy 4.x** controller on a breakout board (32-bit).
- Built a **custom 12" × 16" × 6" enclosure**:
  - Frame: **2020 aluminum extrusion**.
  - Five sides: **0.0625" (1/16") aluminum sheet**.
  - Top: **1/8" clear plexiglass** (visibility into the electronics).

## Connectivity

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The Teensy 4.x supports Ethernet, and the breakout board has the **Ethernet option installed**. Both links to the Teensy are brought out as **panel patch-through connectors on the side of the enclosure**: a **female USB-A** jack and a **female Ethernet (RJ45)** jack.

## Power distribution

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A single **220 V, 20 A branch circuit** feeds the machine through a **front-panel emergency-stop button** (the E-stop breaks the 220 V input, so it drops both the PSU and the spindle drive). Building power arrives as **12/3 from the breaker panel to a junction box**, then via an **extension-cord pigtail** to the panel C20 and the E-stop input; a second pigtail carries the E-stop output to the VFD. This is **single-phase 220 V** (residential); the VFD generates the 3-phase output to the spindle. The 220 V powers an **800 W 48 V Meanwell power supply** and the **VFD**. Everything else is derived from the 48 V rail through two buck converters.

**220 V, 20 A** → front-panel E-stop →

- └ 48 V Meanwell power supply
- └ VFD (spindle drive)

**48 V** (PSU output) →

- └ 4 × stepper motors
- └ 12 V / 3 A buck converter
- └ 5 V / 3 A buck converter

**12 V** (buck) →

- └ 4 × proximity limit sensors
- └ 12 V fan cooling the spindle coolant loop (runs while the spindle runs)
- └ grblHAL Teensy 4.x breakout board, 12 V input — the board uses 12 V to generate the **0–10 V PWM spindle-speed signal** to the VFD

**5 V** (buck) →

- └ breakout board 5 V input — board logic power, **separate from the Teensy** (the Teensy is powered over USB)
- └ 3D probe

## Relay outputs & air assist

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The breakout board provides **5 relay outputs**, each **jumper-selectable for 5 V or 12 V**. Only the **Mist** output is in use, driving the air-assist nozzle:

Mist relay → 12 V **NC** solenoid valve → regulator @ 30 PSI → 1/4" tube (up the umbilical) → needle valve at top of Z gantry → 1/8" copper airline → swivel nozzle aimed where the tool meets the stock

## Connectors & terminations

**Ferrules are crimped on all terminal-block connections** — every stranded conductor landing on a terminal block is ferruled for a clean, captured, reliable termination.

Enclosure pass-throughs use **Molex Mini-Fit Jr panel-mount connectors**. The choice was deliberate: with roughly 60 terminations to make, the builder had far more confidence producing **~60 quality crimps** than ~60 quality solder joints on aircraft-style connectors.

## Cable & wiring schedule

All runs use shielded cable.

| Cable                | Run                                                     | Conductors                                       |
|----------------------|---------------------------------------------------------|--------------------------------------------------|
| <b>18/6 shielded</b> | enclosure → each stepper motor                          | 6 (assignment depends on motor type — see below) |
| <b>22/3 shielded</b> | enclosure → Y-axis proximity switches                   | 12V , Sig , GND                                  |
| <b>22/6 shielded</b> | enclosure → X & Z proximity sensors + 3D probe (shared) | 12V , GND , XLim , ZLim , P , 5V                 |
| <b>12/3</b>          | VFD → spindle (3-phase)                                 | U , V , W                                        |
| <b>22/4 shielded</b> | breakout → VFD (control)                                | 0–10V , run, GND, spare                          |

## 18/6 stepper cable — conductor assignment

| Conductor | Closed-loop stepper | Discrete (open-loop) stepper |
|-----------|---------------------|------------------------------|
| 1         | 48V                 | A+                           |
| 2         | GND                 | A-                           |
| 3         | DIR                 | B+                           |
| 4         | PUL                 | B-                           |
| 5         | EN                  | EN                           |
| 6         | AL                  | AL                           |

**All four axes on this build are closed-loop**, so they all use the left-column assignment ( 48V / GND / DIR / PUL / EN / AL ). DIR / PUL are direction/step to the integrated drives; EN = enable, AL = alarm. The discrete A± / B± column is kept for reference only — the same 18/6 cable would serve an open-loop motor if one were ever fitted. At the enclosure these device cables collapse onto a small set of keyed panel connectors — see [Back-of-enclosure harness & connectors](#).

## Drag chains (cable carriers)

Each axis's drag chain carries every cable that has to continue past it, so loading is cumulative toward the base of the motion chain. **Y is the fullest** — everything routes through the Y carrier first to reach the gantry, then the X carrier feeds the spindle carriage, while the opposite gantry rail (**A**) carries only its own stepper. The two limit-sensor drops that don't need to flex (A and Y) **begin after** their drag chains rather than riding through them.

| Drag chain                                     | Cables carried                                                                                                                                   |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A axis</b><br>(opposite gantry rail)        | 1 × 18/6 (A stepper)<br><i>A limit (22/3) picks up after the chain</i>                                                                           |
| <b>X axis</b><br>(spindle carriage)            | 1 × 12/3 (spindle power)<br>2 × 18/6 (X & Z steppers)<br>1 × 22/6 (X & Z limits + 3D probe)                                                      |
| <b>Y axis</b><br>(gantry — carries everything) | 1 × 12/3 (spindle power)<br>3 × 18/6 (X, Z & Y steppers)<br>1 × 22/6 (X & Z limits + 3D probe)<br><i>Y limit (22/3) picks up after the chain</i> |

That Y bundle — one 12/3, three 18/6, and one 22/6 — would not fit the stock **1.5** drag chain, so the Y axis was moved up to the **next size larger (2")** carrier.

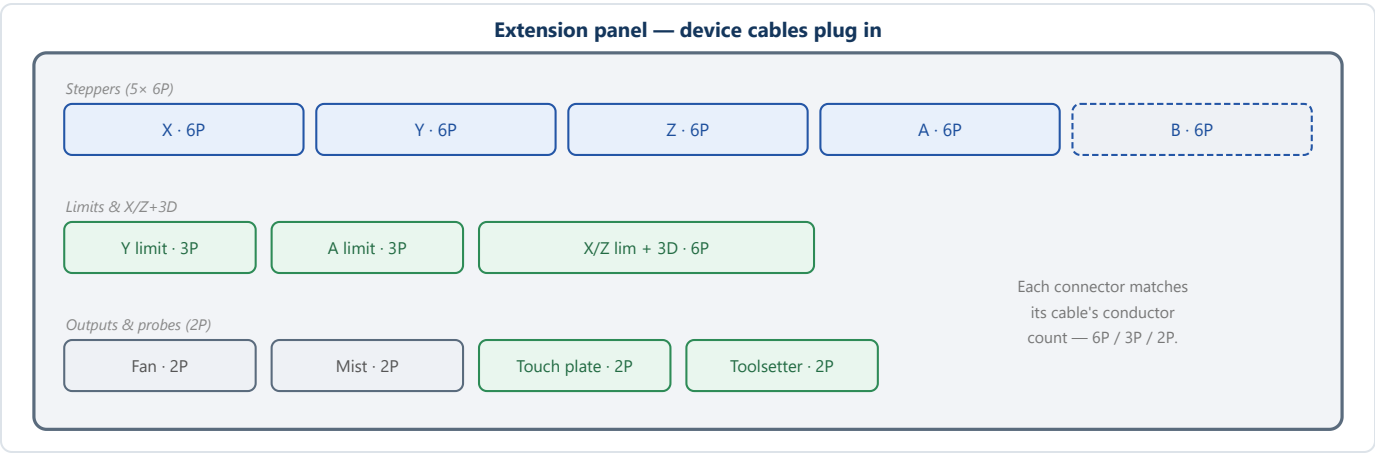
## Back-of-enclosure harness & connectors

The wiring uses **two panels**. The **extension panel** is where the machine's device cables plug in — one socket per cable, sized to its conductor count. The **back panel** on the rear of the enclosure carries the collapsed result to the controller. Because the enclosure lives in a **cabinet drawer** and the back panel is connected **by feel** (drawer pulled forward, hand reaching in), the harness is squeezed down to just **five single-row 1×6 Mini-Fit Jr sockets** there — few connectors, and a linear pin row is easy to count and seat blind.

### Extension panel — device cables plug in

Twelve sockets, one per device cable (50 conductors in):

| Socket                    | Size  | Cable | Conductors                      |
|---------------------------|-------|-------|---------------------------------|
| Stepper X / Y / Z / A / B | 5× 6P | 18/6  | 48V · GND · DIR · PUL · EN · AL |
| Y limit                   | 3P    | 22/3  | 12V · Sig · GND                 |
| A limit                   | 3P    | 22/3  | 12V · Sig · GND                 |
| X/Z limits + 3D probe     | 6P    | 22/6  | 12V · GND · XL · ZL · 3D · 5V   |
| Fan                       | 2P    | —     | 12V · GND                       |
| Mist                      | 2P    | —     | Mist · GND                      |
| Touch plate               | 2P    | —     | TP · GND                        |
| Toolsetter                | 2P    | —     | TS · GND                        |



### Behind the panels — the remap

Between the two panels, each incoming conductor is routed to its designated back-panel socket and pin. The signals that tie together — the **common ground**, the five **AL** **alarms wire-OR'd to one**

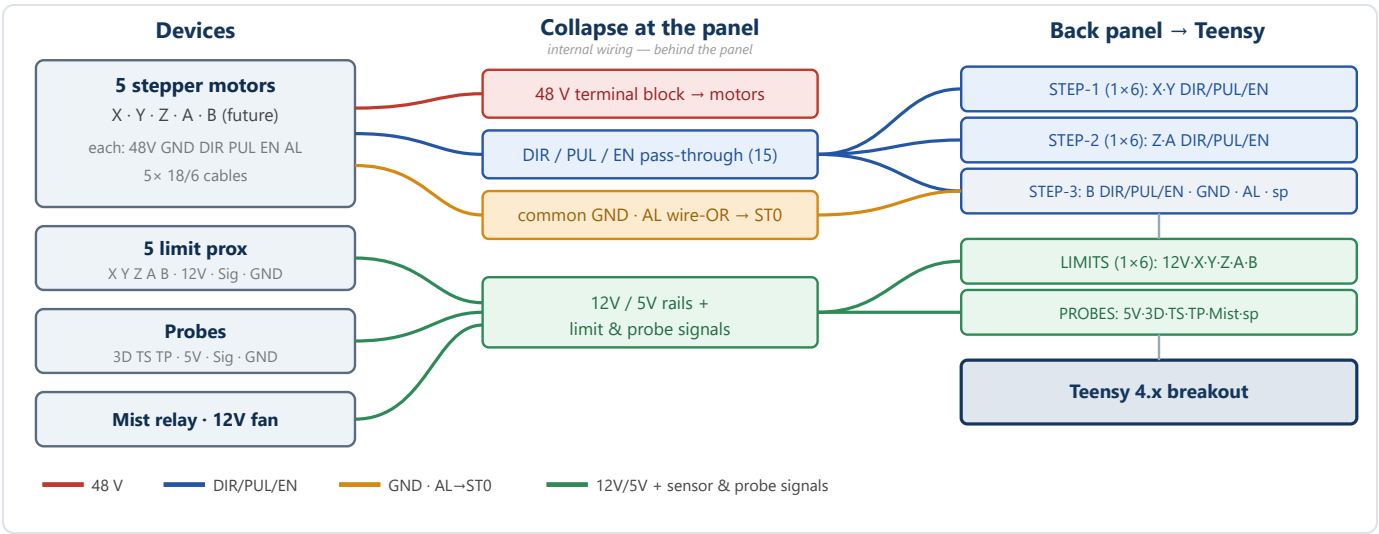
**ST0** , and the shared **12 V / 5 V** rails — are joined by **daisy-chaining from the incoming crimp terminals** (a jumper from one crimp to the next, not a bus bar). Each motor's **48 V splits off to a distribution terminal block**; the spindle **VFD has its own C20 220 V inlet** and a **22/4 control cable** from the breakout (0–10 V speed + run).

Back panel — to the controller

The collapse: a closed-loop motor only needs **DIR, PUL, EN** unique (48 V is its own rail, GND is shared, AL is wire-OR'd), so the 5 stepper channels go from 30 conductors to **5×3 = 15 + GND + AL = 17** → **three 1×6 sockets** (one spare pin). The sensors and probes fill **two more 1×6 sockets**. Net: **five sockets / five cables** to the controller (three 18/6, two 22/6); the **48 V terminal block and common-ground bus are behind the panel**, not on the connect-by-feel face.

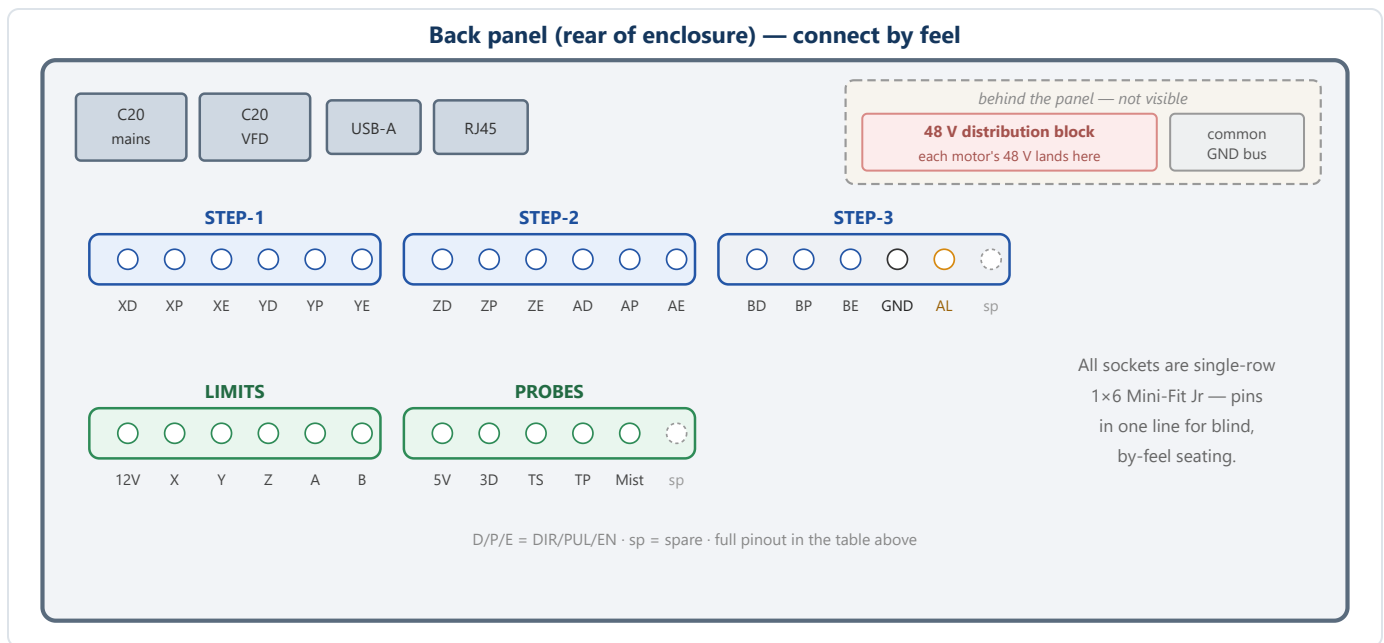
| Socket (1×6) | Cable | 1     | 2     | 3    | 4     | 5     | 6     |
|--------------|-------|-------|-------|------|-------|-------|-------|
| Stepper 1    | 18/6  | X DIR | X PUL | X EN | Y DIR | Y PUL | Y EN  |
| Stepper 2    | 18/6  | Z DIR | Z PUL | Z EN | A DIR | A PUL | A EN  |
| Stepper 3    | 18/6  | B DIR | B PUL | B EN | GND   | AL    | spare |
| Limits       | 22/6  | 12V   | X     | Y    | Z     | A     | B     |
| Probes       | 22/6  | 5V    | 3D    | TS   | TP    | Mist  | spare |

**B** rows are the provisioned 5th-axis / rotary lines. **3D** = 3D probe, **TS** = toolsetter, **TP** = touch plate. The 12 V fan is fed from the panel, not the probes socket; each motor's 48 V and the common ground are distributed separately.



Harness collapse: many device wires → shared GND, wire-OR'd alarm and a 48 V block → five 1×6 sockets into the controller.





Back-panel layout: the five 1×6 sockets plus the 48 V block, common-ground bus and the USB/Ethernet/C20 pass-throughs.

## Overhead umbilical & dust collection

The spindle's services arrive from **above** rather than through a drag chain, on a **ceiling-mounted spring retraction reel** that takes up slack as the gantry moves.

- **Umbilical:** the two 8 mm coolant tubes and a 1/4" compressed-air tube run from the top of the Z gantry into a **1" cable sleeve**.
- **Retraction reel:** the sleeve hangs from a spring-loaded retraction reel on the ceiling above the machine. A **3D-printed cradle** holds the sleeve and clips to the reel hook, so the reel's pull-wire can't wrap and choke the sleeve while still allowing free movement.
- **Routing:** the sleeve terminates at the **back of the CNC**, where the coolant tubes join the pump outlet and the radiator inlet (radiator outlet returns to the reservoir). The air tube comes from the pressure-regulator outlet, whose inlet is fed by the 12 V NC Mist solenoid.
- **Dust collection:** alongside the umbilical, a **4" duct** ends in a **4" swivel** feeding **5' of 4" flex hose** to a **dust shoe** on the spindle.

## Rack-and-pinion drive — setup & maintenance

X (and the Y/A gantry drives) run on rack and pinion. After ~10 hours the X pinion crept **left** under load. A closed-loop motor can't see slip *downstream* of its shaft encoder, so a pinion loosening on the shaft is invisible until a fixed reference (the toolsetter at G59.3) exposes it. With the bores reamed

6→8 mm the set-screw boss is thin and grub-screw grip is the weak point; X slips first because it takes the most direct cutting side-load, but the other reamed pinions carry the same risk.

### Pinion-to-shaft — stop the slip

1. Pull the pinion; confirm the grub screw seats on the shaft **flat** (not a round section), and check whether the screw marred the flat or only kissed it.
2. **Degrease** shaft, bore, and screw threads — oil defeats both grip and threadlocker.
3. Use the **longest grub screw** the thinned boss will take, on the flat; add a second screw if there's room.
4. **Threadlocker:** blue (242/243) is normally enough; with the reduced thread engagement here, **red (271)** is justified as insurance against back-out — just note removal then needs heat (~250 °C, easy on a bare pinion).
5. Torque to screw size (M3 ~1.0–1.4, M4 ~2.5–3, M5 ~5–6 N·m); snug, don't strip the hex. Let red cure ~24 h before heavy cuts.

### Pinion-to-rack mesh & backlash

1. Loosen the motor mount so the pinion can move toward / away from the rack.
2. Shim **0.05–0.10 mm** (feeler gauge, or ~0.1 mm folded paper) between a pinion tooth and the rack, push the pinion in against the shim, tighten the mount, remove the shim.
3. Check the mesh over the **full travel** — it should stay lightly meshed everywhere. Tight at one end + slack at the other means the rack isn't parallel to the axis; realign the rack first.
4. Confirm full tooth-width engagement (pinion not riding a tooth edge).

### Verify

1. By hand: smooth full travel, minimal lash; rock the gantry and feel for play at the pinion.
2. Dial indicator: commanded vs actual on small moves; reverse direction to read backlash (a few thou).
3. Re-run a heavy back-and-forth / cut loop with paint-pen **witness marks** (motor shaft ↔ pinion, and a pinion tooth ↔ a rack tooth) — no separation = fixed.
4. Confirm the **AL → ST0** alarm actually halts grbHAL (force a stall) so a real fault can't pass silently.

### If it still creeps

A single grub screw — especially in a reamed, thin-wall boss — is marginal for rack-and-pinion loads. Durable fixes, in order: **proper 8 mm-bore pinions** (full set-screw boss) → **clamp / split-hub pinion** (grips the whole shaft, no set screw) → **keyed shaft**. Given the reamed pinions, the **clamp-hub style is the recommended upgrade** — it ends set-screw slip for good.

## Sourcing a replacement pinion

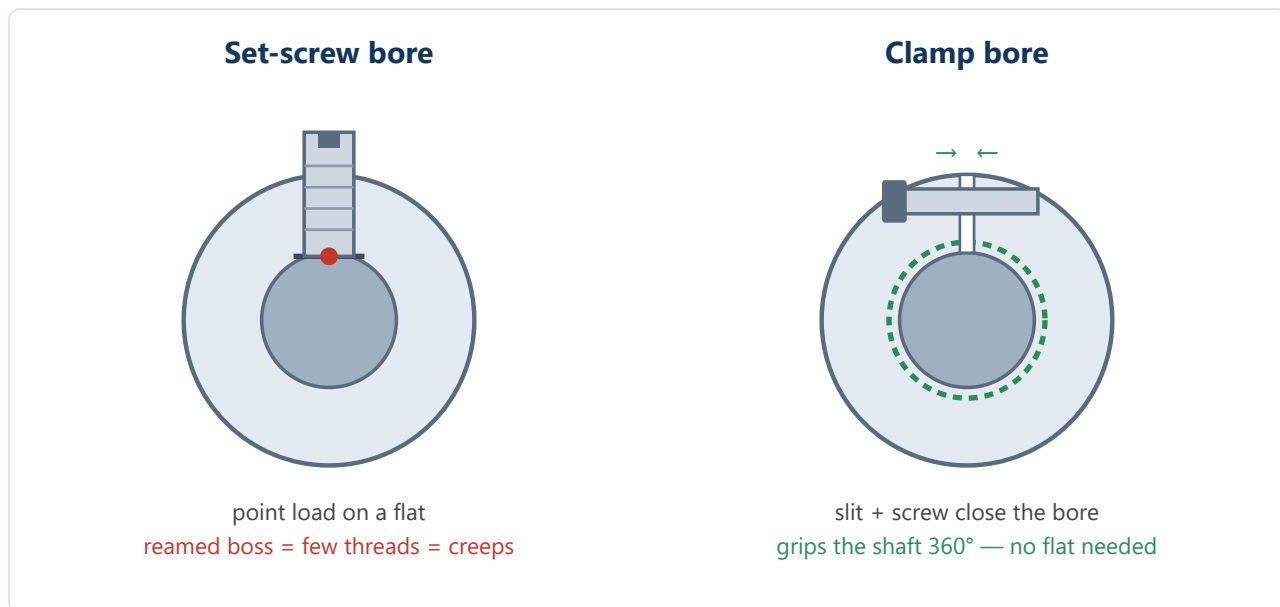
Match the rack on three things, plus the bore:

- **Module** (tooth size) —  $\text{module} = \text{tooth pitch} \div \pi$  (pitch 3.14 mm = Mod 1, 4.71 mm = Mod 1.5, 6.28 mm = Mod 2). Must match the rack exactly.
- **Pressure angle** — almost always **20°**.
- **Tooth count** — match the original to keep steps/mm, or pick a new count and recalibrate **\$100** (X steps/mm) in grblHAL.
- **Bore** — 8 mm, ideally a **clamp or keyed** bore, not a bare set-screw boss.

Sources that let you specify *module + 20° + 8 mm clamp/keyed bore*: **KHK / QTC** (khkgears.us, qtcgears.com — best for CNC rack & pinion), **SDP/SI** (sdp-si.com), **Misumi** (configurable bore & keyway), and **McMaster-Carr** (quick metric-module steel gears). Budget CNC rack-and-pinion sets are on Amazon / AliExpress (Mod 1/1.5, 20T, often 8 mm bore) with variable quality.

## Clamp hub vs set-screw hub

A **clamp (split) hub** grips like a clamp-style shaft collar: a slit is sawn through the bore and a screw bridges it; tightening the screw squeezes the slit closed so the **whole bore clamps the shaft 360°** — far more holding torque, no marring, and no dependence on a thin set-screw boss (exactly the reamed-pinion weakness).



Two routes to a clamp mount: a pinion with an **integral clamp bore** (specify it when ordering from KHK / Misumi / SDP), or a **keyless locking bushing** (Trantorque GT, Fenner B-LOC, ...) dropped inside a larger-bore pinion — the bushing expands to grip both the shaft and the bore, letting you use almost any pinion of the right module on the 8 mm shaft. The latter is handy here, since matching bore *and* gearing was the sticking point.

## Bill of materials

| Item                                                               | Qty | Unit price | Extended | Source                            |
|--------------------------------------------------------------------|-----|------------|----------|-----------------------------------|
| <b>Machine &amp; frame</b>                                         |     |            |          |                                   |
| Mega V XL (Kickstarter edition) — base machine                     | 0   | \$0.00     | \$0.00   |                                   |
| Mark Smith riser kit (+3" Z clearance)                             | 0   | \$0.00     | \$0.00   |                                   |
| Mark Smith Z gantry assembly                                       | 0   | \$0.00     | \$0.00   |                                   |
| <b>Controller &amp; I/O</b>                                        |     |            |          |                                   |
| Teensy 4.1 SOM (grblHAL controller)                                | 0   | \$0.00     | \$0.00   |                                   |
| Teensy 4.x breakout board (Ethernet option)                        | 0   | \$0.00     | \$0.00   |                                   |
| USB-A pass-through, panel mount                                    | 0   | \$0.00     | \$0.00   |                                   |
| Ethernet pass-through, panel mount                                 | 0   | \$0.00     | \$0.00   |                                   |
| E-stop button (breaks 220 V)                                       | 0   | \$0.00     | \$0.00   |                                   |
| <b>Power</b>                                                       |     |            |          |                                   |
| Meanwell 48 V power supply, 800 W                                  | 0   | \$0.00     | \$0.00   |                                   |
| C20 panel inlet — mains 220 V                                      | 0   | \$0.00     | \$0.00   |                                   |
| C20 panel inlet — VFD 220 V                                        | 0   | \$0.00     | \$0.00   |                                   |
| Power relay accessory (110 V pump relay)                           | 0   | \$0.00     | \$0.00   | <a href="#">Amazon B07Z6R1KP3</a> |
| <b>Spindle &amp; cooling</b>                                       |     |            |          |                                   |
| 2.2 kW 80 mm water-cooled spindle + 2.2 kW VFD kit (ER20, 24k RPM) | 0   | \$0.00     | \$0.00   | <a href="#">Amazon B073TRP92V</a> |
| Braking resistor, 150 W                                            | 0   | \$0.00     | \$0.00   |                                   |
| Submersible pump, 110 V                                            | 0   | \$0.00     | \$0.00   |                                   |
| Coolant reservoir, 1 gal                                           | 0   | \$0.00     | \$0.00   |                                   |
| Radiator with integrated 12 V fan(s)                               | 0   | \$0.00     | \$0.00   |                                   |
| 8 mm clear tubing (25')                                            | 0   | \$0.00     | \$0.00   |                                   |
| Coolant mix — antifreeze + distilled water + bleach                | 0   | \$0.00     | \$0.00   |                                   |

| Item                                                                     | Qty | Unit price | Extended | Source |
|--------------------------------------------------------------------------|-----|------------|----------|--------|
| <b>Motion</b>                                                            |     |            |          |        |
| Stepper, 1.2 N·m (Z)                                                     | 0   | \$0.00     | \$0.00   |        |
| Stepper, 2.0 N·m (X)                                                     | 0   | \$0.00     | \$0.00   |        |
| Stepper, 3.2 N·m (Y / A)                                                 | 0   | \$0.00     | \$0.00   |        |
| Drag chain, 2" (Y-axis upsize)                                           | 0   | \$0.00     | \$0.00   |        |
| <b>Sensing &amp; probing</b>                                             |     |            |          |        |
| Proximity sensors (limits / homing)                                      | 0   | \$0.00     | \$0.00   |        |
| 3D mechanical probe                                                      | 0   | \$0.00     | \$0.00   |        |
| Toolsetter probe                                                         | 0   | \$0.00     | \$0.00   |        |
| <b>Air assist</b>                                                        |     |            |          |        |
| 1/4" black tubing (10')                                                  | 0   | \$0.00     | \$0.00   |        |
| Needle valve                                                             | 0   | \$0.00     | \$0.00   |        |
| Copper tubing (12")                                                      | 0   | \$0.00     | \$0.00   |        |
| Swivel nozzle, 1/8"                                                      | 0   | \$0.00     | \$0.00   |        |
| Solenoid valve, 12 V NC                                                  | 0   | \$0.00     | \$0.00   |        |
| Pressure regulator (30 PSI)                                              | 0   | \$0.00     | \$0.00   |        |
| 3D-printed nozzle bracket                                                | 0   | \$0.00     | \$0.00   |        |
| <b>Cabling &amp; connectors</b>                                          |     |            |          |        |
| Shielded cable, 18/6 (100')                                              | 0   | \$0.00     | \$0.00   |        |
| Shielded cable, 22/6 (50')                                               | 0   | \$0.00     | \$0.00   |        |
| Shielded cable, 22/4 (10')                                               | 0   | \$0.00     | \$0.00   |        |
| Shielded cable, 22/3 (25')                                               | 0   | \$0.00     | \$0.00   |        |
| 12/3 cable — mains feed (breaker panel → junction box)                   | 0   | \$0.00     | \$0.00   |        |
| 12/3 extension-cord pigtail — E-stop power (junction→E-stop, E-stop→VFD) | 0   | \$0.00     | \$0.00   |        |

| Item                                                             | Qty | Unit price | Extended      | Source |
|------------------------------------------------------------------|-----|------------|---------------|--------|
| 12/3 cable — VFD → spindle, 3-phase (U/V/W, through drag chains) | 0   | \$0.00     | \$0.00        |        |
| Mini-Fit Jr 1×6 connectors (5 mated pairs)                       | 0   | \$0.00     | \$0.00        |        |
| Mini-Fit Jr crimp terminals (~60)                                | 0   | \$0.00     | \$0.00        |        |
| 48 V distribution terminal block                                 | 0   | \$0.00     | \$0.00        |        |
| <b>Total</b>                                                     |     |            | <b>\$0.00</b> |        |

**Not yet itemized** (described in the build above but missing from this BOM):

- **Enclosure** — 2020 extrusion, 1/16" aluminum sheet (5 sides), 1/8" plexiglass top.
- **Buck converters** — 12 V 3 A and 5 V 3 A.
- **Ferrules** for the terminal-block connections.
- **Touch plate** (probe input in parallel with the 3D probe).
- **Dust collection** — 4" duct, 4" swivel, 5' of 4" flex hose, dust shoe.
- **Overhead service rig** — spring retraction reel, 1" cable sleeve, 3D-printed sleeve cradle.